



SHORE

Examination Number:

Set:

Year 12

Mathematics

Trial HSC Examination

August 2018

General instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”
- A NESA reference sheet is provided

Total marks – 100

Section I

Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 6 – 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which expression is a factorisation of $27x^3 - 8$?

A. $(3x - 2)(9x^2 - 6x + 4)$

B. $(3x - 2)(9x^2 + 6x + 4)$

C. $(3x - 2)(9x^2 - 12x + 4)$

D. $(3x - 2)(9x^2 + 12x + 4)$

2 The first three terms of an arithmetic series are 4, 7 and 10.
What is the sum of the first 18 terms?

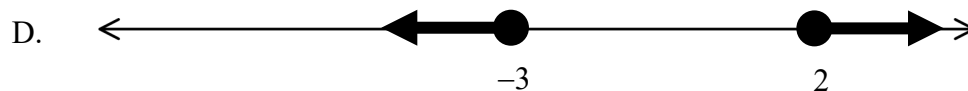
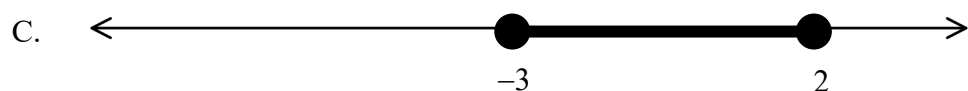
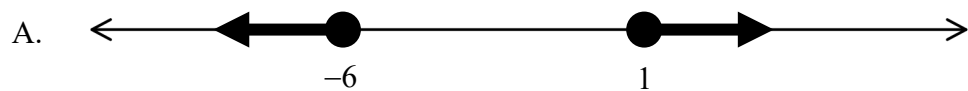
A. 55

B. 495

C. 531

D. 558

3 What is the solution to $x^2 + 5x - 6 \geq 0$ when graphed on the number line?



- 4 The following two statements are made about the function $f(x) = \frac{3x^3}{x-2}$.

Statement 1: The function is discontinuous at $x = 2$.

Statement 2: The function is an odd function.

Which of the following is true?

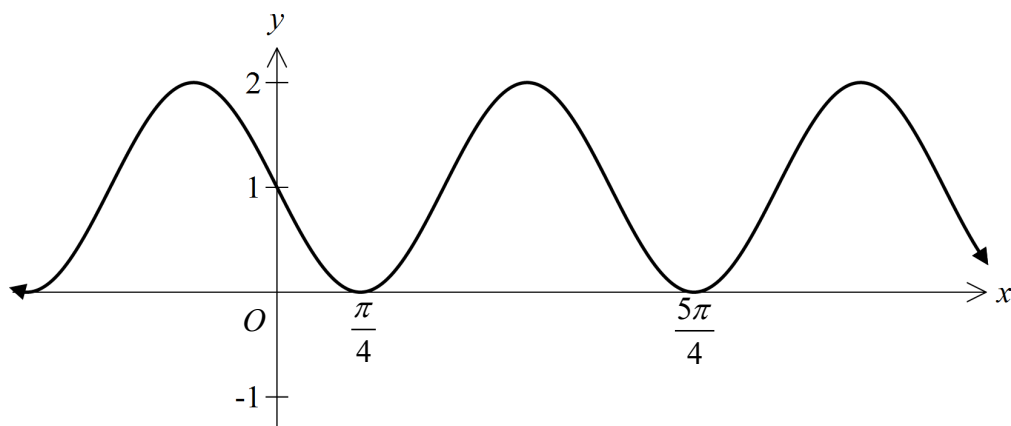
- A. Statement 1 and Statement 2 are both correct.
 - B. Statement 1 is correct and Statement 2 is incorrect.
 - C. Statement 1 is incorrect and Statement 2 is correct.
 - D. Statement 1 and statement 2 are both incorrect.
- 5 Using Simpson's rule with 5 function values, which expression gives you the area under the curve $y = x \ln x$ between $x = 2$ and $x = 4$?

- A. $\frac{1}{4}(2 \ln 2 + 5 \ln 2.5 + 6 \ln 3 + 7 \ln 3.5 + 4 \ln 4)$
- B. $\frac{1}{6}(2 \ln 2 + 5 \ln 2.5 + 6 \ln 3 + 7 \ln 3.5 + 4 \ln 4)$
- C. $\frac{1}{4}(2 \ln 2 + 10 \ln 2.5 + 6 \ln 3 + 14 \ln 3.5 + 4 \ln 4)$
- D. $\frac{1}{6}(2 \ln 2 + 10 \ln 2.5 + 6 \ln 3 + 14 \ln 3.5 + 4 \ln 4)$

- 6 For a particular angle, θ , $\cos \theta = -\frac{3}{\sqrt{34}}$ and $\tan \theta < 0$. What is the value of $\operatorname{cosec} \theta$?

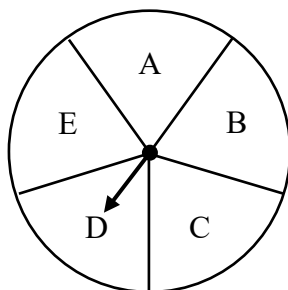
- A. $\frac{5}{3}$
- B. $-\frac{\sqrt{34}}{5}$
- C. $\frac{\sqrt{34}}{5}$
- D. $-\frac{3}{5}$

- 7 The curve below is of the form $y = 1 + a \sin(bx)$ where a and b are constants.



What are the values of a and b ?

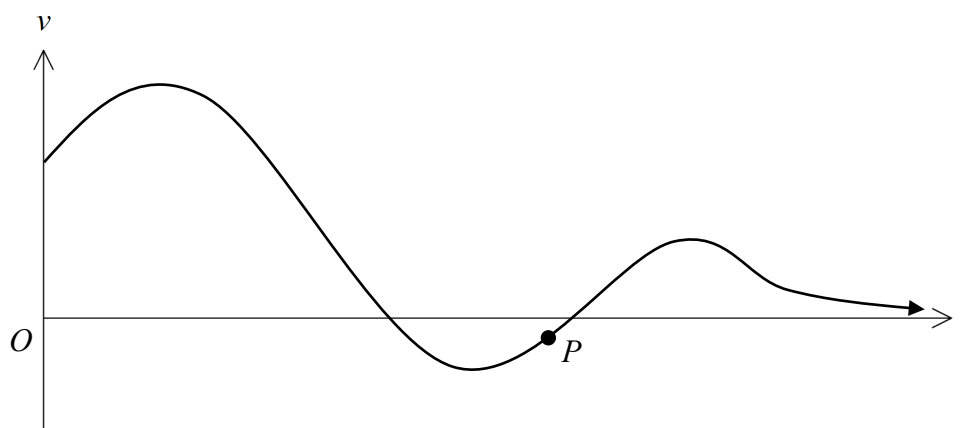
- A. $a = 1$ and $b = 2$
 - B. $a = -1$ and $b = 2$
 - C. $a = -2$ and $b = 4$
 - D. $a = -1$ and $b = 4$
- 8 A spinner is marked with the letter A, B, C, D and E. When it is spun, each of the five letters is equally likely to occur.



The spinner is spun three times. What is the probability that a consonant appears on at least one of the three spins?

- A. $\frac{8}{125}$
- B. $\frac{27}{125}$
- C. $\frac{3}{5}$
- D. $\frac{117}{125}$

- 9 A particle initially starts at the origin. The graph, drawn to scale below, shows the velocity, v , of the particle moving along a straight line as a function of time, t .



Which statement describes the motion of the particle at point P ?

- A. It is to the left of the origin with positive acceleration.
 - B. It is to the left of the origin with negative acceleration.
 - C. It is to the right of the origin with positive acceleration.
 - D. It is to the right of the origin with negative acceleration.
- 10 The roots of the equation $px^2 + qx + r = 0$ are α and β .
What is the simplified expression for $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$?

- A. $\frac{q^2 - 2pr}{r^2}$
- B. $\frac{q^2 - 2p}{r}$
- C. $\frac{q^2 - pr}{r^2}$
- D. $\frac{r^2 - 2pq}{q^2}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

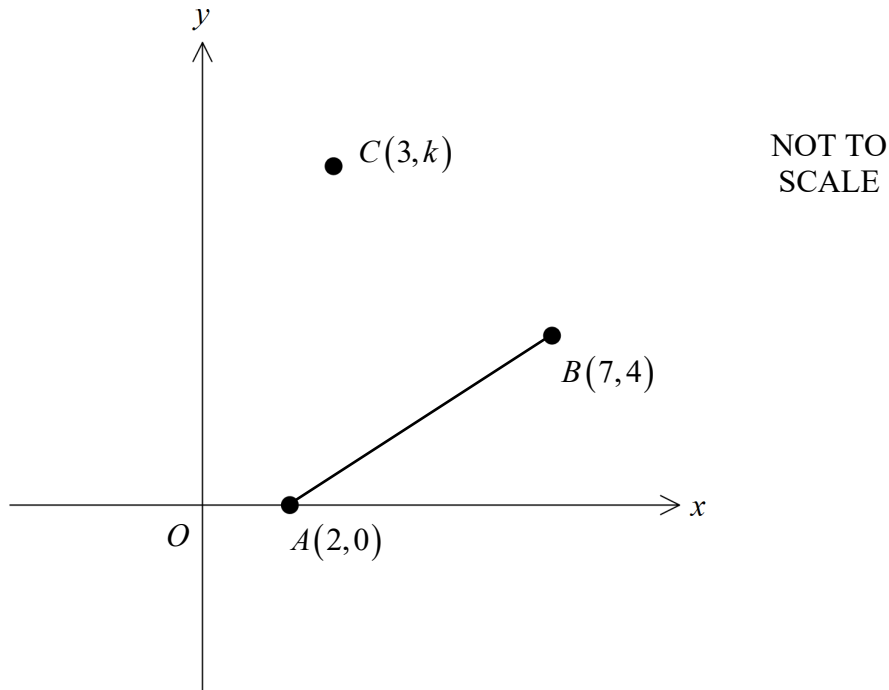
Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Simplify $3x - 4x(2 + 5y)$. 1
- (b) Express $\frac{\sqrt{6}}{5 - \sqrt{3}}$ with a rational denominator. 2
- (c) Differentiate $(\cos x + 2)^5$. 2
- (d) Differentiate $\frac{e^{3x}}{2x}$. 2
- (e) Find $\int (1 + \sec^2 3x) dx$. 2
- (f) A parabola has equation $(x - 2)^2 = 20(y + 3)$. Find the coordinates of the focus. 2
- (g) Find the limiting sum of the infinite geometric series $\frac{7}{3} + \frac{14}{9} + \frac{28}{27} + \dots$. 2
- (h) Write down the domain and range of the function $y = \frac{1}{\sqrt{9 - x}}$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Find the gradient of the normal to the curve $y = \log_e(x^2 - 3)$ at the point $(2, 0)$. **3**

(b) The points $A(2, 0)$, $B(7, 4)$ and $C(3, k)$ lie on a number plane.



(i) Show that the equation of line AB is $4x - 5y - 8 = 0$. **2**

The point C has coordinates $(3, k)$, where $k > 0$ and the perpendicular distance from C to AB is $\sqrt{41}$.

(ii) Show k can be found using the equation $41 = |4 - 5k|$ and hence find the value of k . **3**

(iii) Prove triangle ABC is right angled. **2**

(c) Evaluate $\int_1^8 \frac{1}{\sqrt{x}} dx$, leaving your answer in the form $a + b\sqrt{2}$, **3**
where a and b are integers.

(d) Sketch the region defined by $y \geq \frac{2}{x+1}$. **2**

Question 13 (15 marks) Use a SEPARATE writing booklet

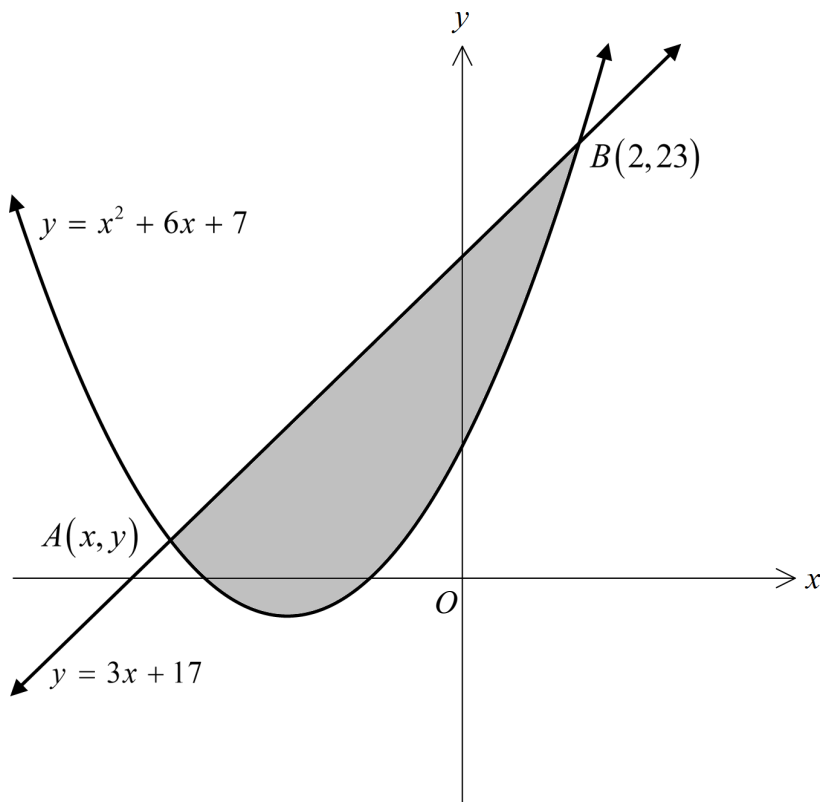
(a) Find $\int \frac{x}{x^2 + 3} dx$. 2

(b) Points A and B have coordinates $(4, -1)$ and $(12, 11)$ respectively.

(i) Find the coordinates of midpoint AB . 1

(ii) Given AB is the diameter of a circle, find the equation of the circle. 2

(c) The line $y = 3x + 17$ intersects the curve $y = x^2 + 6x + 7$ at the points $A(x, y)$ and $B(2, 23)$.



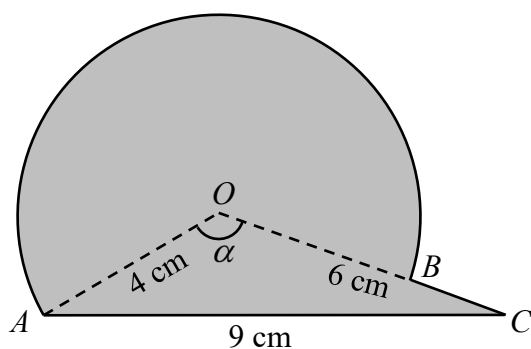
(i) Find the x coordinate of point A . 2

(ii) Find the exact area bounded by the line and the curve. 3

Question 13 continues on page 9

Question 13 (continued)

- (d) In the figure below $OC = 6$ cm, $AC = 9$ cm, $AO = 4$ cm and $\angle AOB = \alpha$.



- | | | |
|------|--|----------|
| (i) | Show that, correct to 2 decimal places, $\alpha = 2.22$ radians. | 2 |
| (ii) | Find the area of the shade shape, correct to 1 decimal place. | 3 |

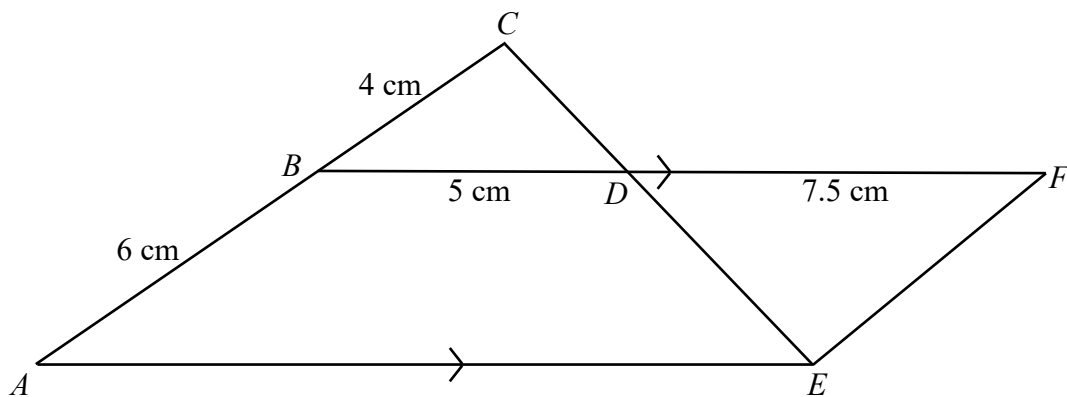
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) Consider the curve $y = -x^3 + 3x^2 - 4$.

- (i) Find the coordinates of the stationary points and determine their nature. 3
- (ii) Find the coordinates of any point of inflexion. 2
- (iii) Sketch the curve, labelling the y -intercept, stationary points and any point of inflexion. Do not find the x -intercepts. 2

(b) In the diagram below $BF \parallel AE$, $AB = 6 \text{ cm}$, $BC = 4 \text{ cm}$, $BD = 5 \text{ cm}$ and $DF = 7.5 \text{ cm}$.



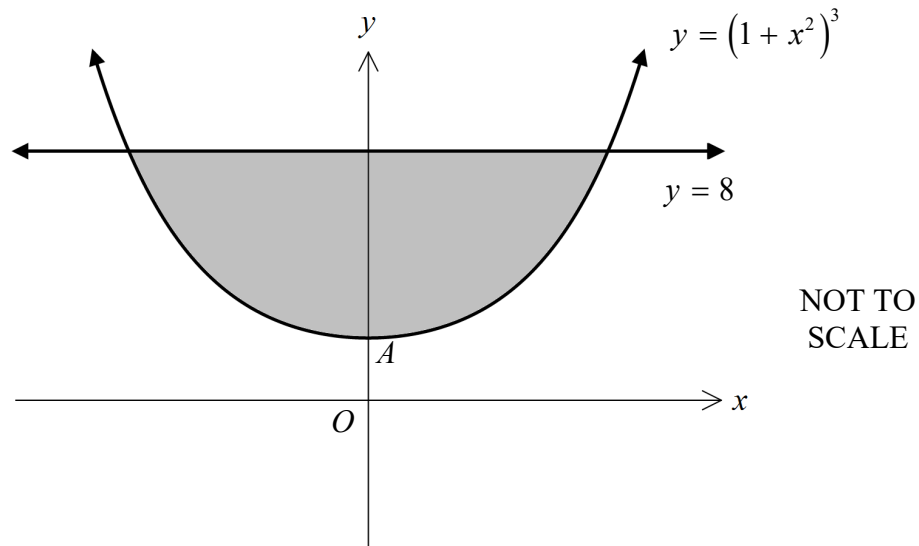
- (i) Show that $DE = \frac{3CD}{2}$, giving reasons. 2
 - (ii) Given $CD = 2 \text{ cm}$, prove $\triangle BCD \parallel \triangle FED$. 2
- (c) A company predicts a yearly profit of \$140 000 in the year 2018. The company predicts that the yearly profit will rise each year by 5%.
- (i) Show the predicted profit in the year 2020 is given by $140\,000 \times 1.05^2$. 1
 - (ii) Find the total predicted profit for the years 2018 to 2030 inclusive, giving your answer to the nearest dollar. 3

Question 15 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_1^e \frac{x^2 + 1}{x} dx$. **3**

(b) Solve $2 \cos^2 x - 7 \sin x + 2 = 0$ for $0 \leq x \leq 2\pi$. **3**

- (c) The region enclosed by $y = (1 + x^2)^3$ and the line $y = 8$ is shown below.
The curve $y = (1 + x^2)^3$ intersects the y -axis at point A .



- (i) Find the coordinates of point A . **1**
- (ii) The region is rotated about the y -axis. Find the volume of the solid formed, leaving your answer in terms of π . **3**
- (d) Consider the equation $2 \log_2 (x + 15) - \log_2 x = 6$.
- (i) Show that $x^2 - 34x + 225 = 0$. **3**
- (ii) Hence, or otherwise, solve the equation $2 \log_2 (x + 15) - \log_2 x = 6$. **2**

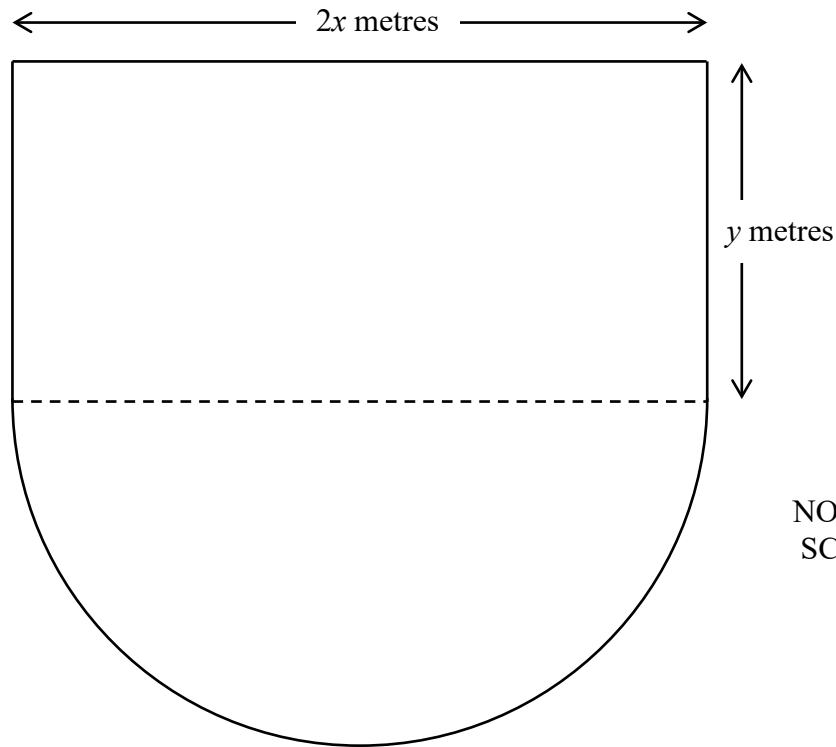
Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) The velocity, $v \text{ ms}^{-1}$, of a particle moving along a straight line is given by $v = 3t^2 - 12t + 14$, where t is the time in seconds.
- (i) Find the initial velocity of the particle. **1**
 - (ii) Show that v is positive for all values of t . **2**
 - (iii) Find the distance travelled between the times $t = 1$ and $t = 3$. **2**
- (b) A scientist is researching the growth of a certain species of hamster. She proposes that the length, $x \text{ cm}$, of a hamster t days after birth is given by $x = 15 - 12e^{-\frac{t}{14}}$.
- (i) Find the length of a hamster when it is born. **1**
 - (ii) Find the number of days it takes for a hamster to grow to 10 cm in length, correct to 1 decimal place. **2**
 - (iii) Find the rate of growth of the hamster 8 days after birth. Give your answer in cm per day, correct to 1 decimal place. **2**

Question 16 continues on page 13

Question 16 (continued)

- (c) A stage, with a perimeter of 80 metres is made by joining a rectangle and a semicircle. The length of the rectangle is $2x$ metres and the width of the rectangle is y metres. The diameter of the semicircle is $2x$ metres.



- (i) Show that the area, $A \text{ m}^2$, of the stage is given by $A = 80x - \left(2 + \frac{\pi}{2}\right)x^2$. **2**
- (ii) Find the value of x , correct to 3 significant figures, that will provide the maximum stage area. Justify why this value of x gives the maximum area. **3**

End of paper

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2018 - TRIAL HSC - MATHEMATICS

① B ② C ③ A ④ B ⑤ D ⑥ C ⑦ B ⑧ D ⑨ C ⑩ A

① $27x^3 - 8 = (3x)^3 - 2^3$
 $= (3x-2)(9x^2 + 6x + 4)$
 $\therefore B$

② 4, 7, 10 $\therefore a=4, d=3$
 $S_{18} = \frac{18}{2} (2(4) + (18-1) \times 3)$
 $= 9 \times 59$
 $= 531$
 $\therefore C$

③ $x^2 + 5x - 6 \geq 0$
 crit. pts $x^2 + 5x - 6 = 0$
 $(x+6)(x-1) = 0$
 $\therefore x = -6 \text{ or } x = 1$



$\therefore x \leq -6 \text{ or } x \geq 1$
 $\therefore A$

④ $f(x) = \frac{3x^3}{x-2}$

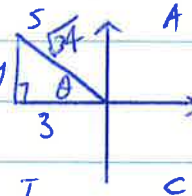
$x \neq 2 \therefore$ statement 1 true
 $f(-x) = \frac{3(-x)^3}{-x-2}$
 $= \frac{-3x^3}{-x-2}$
 $\neq -f(x)$
 $\therefore$ statement 2 is false
 $\therefore B$

⑤ $y = x \ln x$

x	2	2.5	3	3.5	4
y	2ln2	2.5ln2.5	3ln3	3.5ln3.5	4ln4

$h = \frac{1}{2}$
 $A = \frac{\frac{1}{2}}{3} [2\ln 2 + 4(2.5\ln 2.5 + 3.5\ln 3.5) + 2\ln 3 + 4\ln 4]$
 $= \frac{1}{6} [2\ln 2 + 10\ln 2.5 + 6\ln 3 + 14\ln 3.5 + 4\ln 4]$
 $\therefore D$

⑥ $\cos \theta = \frac{3}{\sqrt{34}}, \tan \theta < 0$



$y = \sqrt{(\sqrt{34})^2 - 3^2}$
 $= \sqrt{25}$
 $= 5$
 $\therefore \sin \theta = \frac{5}{\sqrt{34}}$
 $\therefore \operatorname{cosec} \theta = \frac{\sqrt{34}}{5}$
 $\therefore C$

⑦ $y = 1 + a \sin(bx)$
 • flipped with amplitude 1 $\therefore a = -1$
 • repeats after $\pi \therefore b = 2$
 $\therefore B$

⑧ $P(\text{consonant}) = 1 - P(\text{no consonants})$
 $= 1 - \left(\frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}\right)$
 $= \frac{117}{125}$
 $\therefore D$

⑨ negative velocity but still right of origin
 positive acceleration
 $\therefore C$

$$\textcircled{10} \quad \alpha + \beta = -\frac{q}{p} \quad \alpha\beta = \frac{r}{p}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{\frac{q^2}{p^2} - \frac{2r}{p}}{\frac{r^2}{p^2}}$$

$$= \frac{q^2 - 2pr}{p^2} \div \frac{r^2}{p^2}$$

$$= \frac{q^2 - 2pr}{r^2}$$

Question 11

$$\textcircled{ii} \text{ a) } 3x - 4x(2 + 5y) = 3x - 8x - 20xy \\ = -5x - 20xy$$

$$\text{b) } \frac{\sqrt{6}}{5 - \sqrt{3}} = \frac{\sqrt{6}}{5 - \sqrt{3}} \times \frac{5 + \sqrt{3}}{5 + \sqrt{3}}$$

$$= \frac{5\sqrt{6} + \sqrt{18}}{5^2 - (\sqrt{3})^2}$$

$$= \frac{5\sqrt{6} + 3\sqrt{2}}{22}$$

$$\text{c) } \frac{d}{dx} (\cos x + 2)^5 = 5(\cos x + 2)^4 \times -\sin x \\ = -5\sin x (\cos x + 2)^4$$

$$\text{d) } \frac{d}{dx} \frac{e^{3x}}{2x} = \frac{vu' - uv'}{v^2}$$

$$u = e^{3x} \quad v = 2x \\ u' = 3e^{3x} \quad v' = 2$$

$$= \frac{2x \times 3e^{3x} - 2 \times e^{3x}}{(2x)^2}$$

$$= \frac{6xe^{3x} - 2e^{3x}}{4x^2}$$

$$= \frac{2e^{3x}(3x - 1)}{4x^2}$$

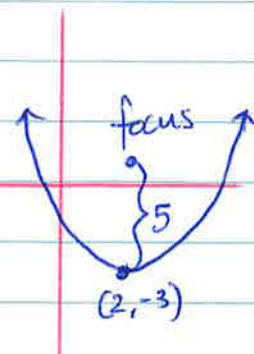
$$= \frac{e^{3x}(3x - 1)}{2x^2}$$

$$\text{e) } \int 1 + \sec^2 3x \, dx = x + \frac{1}{3} \tan 3x + c$$

$$f) (x-2)^2 = 20(y+3)$$

$\therefore$ vertex $(2, -3)$ focal length $= 5$.

$\therefore$ focus $(2, 2)$



$$g) \frac{7}{3} + \frac{14}{9} + \frac{28}{27} + \dots$$

$$a = \frac{7}{3} \quad r = \frac{2}{3}$$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\frac{7}{3}}{1 - \frac{2}{3}}$$

$$= 7$$

$$h) y = \frac{1}{\sqrt{9-x}}$$

domain $x: x < 9$

range $y: y > 0$

Question 12

(12) a) $y = \log_e(x^2 - 3)$

$$y' = \frac{1}{x^2 - 3} \times 2x$$

$$= \frac{2x}{x^2 - 3}$$

at $x = 2$, $y' = \frac{2(2)}{2^2 - 3}$

$$= 4$$

$$\therefore M_{\text{tangent}} = 4$$

$$\therefore M_{\text{normal}} = -\frac{1}{4}$$

b) i) $M_{AB} = \frac{4-0}{7-2}$

$$= \frac{4}{5}$$

$A(2, 0)$ $B(7, 4)$ $C(3, k)$

$$\therefore y - 0 = \frac{4}{5}(x - 2)$$

$$5y = 4(x - 2)$$

$$5y = 4x - 8$$

$$0 = 4x - 5y - 8$$

ii) $d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$

$$\sqrt{41} = \frac{|4(3) + -5(k) + (-8)|}{\sqrt{4^2 + (-5)^2}}$$

$$\sqrt{41} = \frac{|4 - 5k|}{\sqrt{41}}$$

$$41 = |4 - 5k|$$

$$\therefore 41 = |4 - 5k|$$

$$4 - 5k = 41$$

$$-5k = 37$$

$$k = -\frac{37}{5}$$

or

$$4 - 5k = -41$$

$$-5k = -45$$

$$k = 9$$

$$k > 0, \therefore k = 9$$

$$\text{iii) } m_{AB} = \frac{4}{5}$$

$$B(7, 4) \quad C(3, 9)$$

$$m_{BC} = \frac{9-4}{3-7}$$

$$= \frac{5}{-4}$$

$$= -\frac{5}{4}$$

$$m_{AB} \times m_{BC} = \frac{4}{5} \times -\frac{5}{4}$$

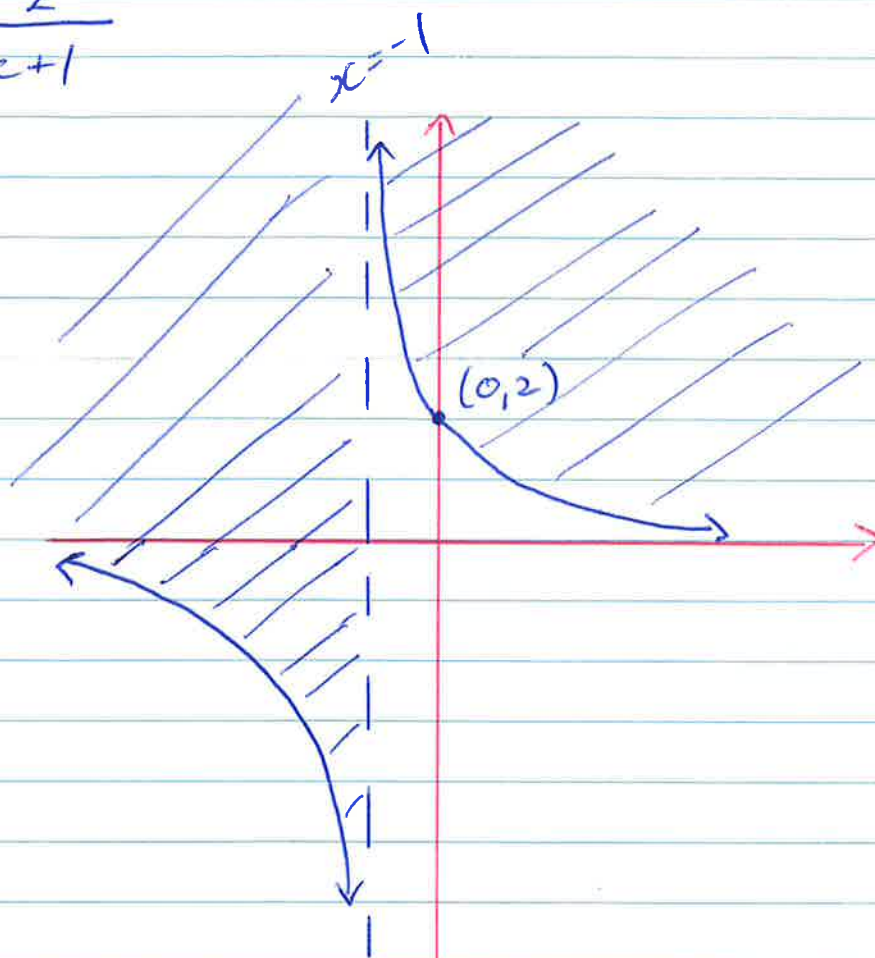
$$= -1$$

$$\therefore AB \perp BC$$

$\therefore \triangle ABC$ is right angled

$$\begin{aligned} \text{c) } \int_1^8 \frac{1}{\sqrt{x}} dx &= \int_1^8 x^{-\frac{1}{2}} dx \\ &= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^8 \\ &= \left[2\sqrt{x} \right]_1^8 \\ &= (2\sqrt{8}) - (2\sqrt{1}) \\ &= 2 \times 2\sqrt{2} - 2 \\ &= -2 + 4\sqrt{2} \end{aligned}$$

d) $y \geq \frac{2}{x+1}$



Question 13

$$\textcircled{13} \text{ a) } \int \frac{x}{x^2+3} dx = \frac{1}{2} \times \int \frac{2x}{x^2+3} dx$$

$$= \frac{1}{2} \log_e(x^2+3) + C$$

$$\text{b) } A(4, -1) \quad B(12, 11)$$

$$\text{i) } MP_{AB} = \left(\frac{4+12}{2}, \frac{-1+11}{2} \right)$$

$$= (8, 5)$$

$$\text{ii) distance } B \text{ to } MP = \sqrt{(11-5)^2 + (12-8)^2}$$

$$= \sqrt{52}$$

$$\therefore (x-8)^2 + (y-5)^2 = (\sqrt{52})^2$$

$$(x-8)^2 + (y-5)^2 = 52$$

$$\text{c) } y = 3x + 17 \quad \& \quad y = x^2 + 6x + 7$$

$$\text{i) } x^2 + 6x + 7 = 3x + 17$$

$$x^2 + 3x - 10 = 0$$

$$(x+5)(x-2) = 0$$

$$\therefore x = -5 \quad \& \quad x = 2$$

$$\therefore x\text{-coordinate of } A \text{ is } -5$$

$$\begin{aligned}
 \text{i) Area} &= \int_{-5}^2 [3x+17 - (x^2+6x+7)] dx \\
 &= \int_{-5}^2 -x^2 - 3x + 10 dx \\
 &= \left[-\frac{x^3}{3} - \frac{3x^2}{2} + 10x \right]_{-5}^2 \\
 &= \left(-\frac{2^3}{3} - \frac{3(2)^2}{2} + 10(2) \right) - \left(-\frac{(-5)^3}{3} - \frac{3(-5)^2}{2} + 10(-5) \right) \\
 &= \frac{34}{3} - \left(-\frac{275}{6} \right) \\
 &= \frac{343}{6} \text{ u}^2 \quad \text{or } 57\frac{1}{6} \text{ u}^2
 \end{aligned}$$

$$d) i) \cos \alpha = \frac{4^2 + 6^2 - 9^2}{2 \times 4 \times 6}$$

$$\cos \alpha = -\frac{29}{48}$$

$$\alpha = \cos^{-1} \left(-\frac{29}{48} \right)$$

$$\alpha = 2.2195..$$

$$\alpha = 2.22$$

$$\begin{aligned}
 \text{ii) Area}_{\Delta} &= \frac{1}{2} \times 4 \times 6 \times \sin 2.22 \\
 &= 9.5587
 \end{aligned}$$

$$\begin{aligned}
 \text{Area}_{\text{sector}} &= \frac{1}{2} \times 4^2 \times (2\pi - 2.22) \\
 &= 32.505..
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area} &= 9.5587.. + 32.505.. \\
 &= 42.1 \text{ cm}^2
 \end{aligned}$$

Question 14

(14) a) $y = -x^3 + 3x^2 - 4$

i) stat pts when $y' = 0$

$$y' = -3x^2 + 6x$$

$$0 = -3x^2 + 6x$$

$$0 = -3x(x-2)$$

$$\therefore x = 0 \quad \text{or} \quad x = 2$$

$$y'' = -6x + 6$$

$$\text{at } x = 0 \quad y'' = -6(0) + 6 = 6$$

$\therefore$ min t.p.

$$\text{at } x = 2 \quad y'' = -6(2) + 6 = -6$$

$\therefore$ max t.p.

$$\text{at } x = 0 \quad y = -(0)^3 + 3(0)^2 - 4 = -4$$

$\therefore (0, -4)$ is min t.p.

$$\text{at } x = 2 \quad y = -2^3 + 3(2)^2 - 4 = 0$$

$\therefore (2, 0)$ is max t.p.

ii) pt of inflexion at $y'' = 0$

$$y'' = -6x + 6$$

$$0 = -6x + 6$$

$$6x = 6$$

$$x = 1$$

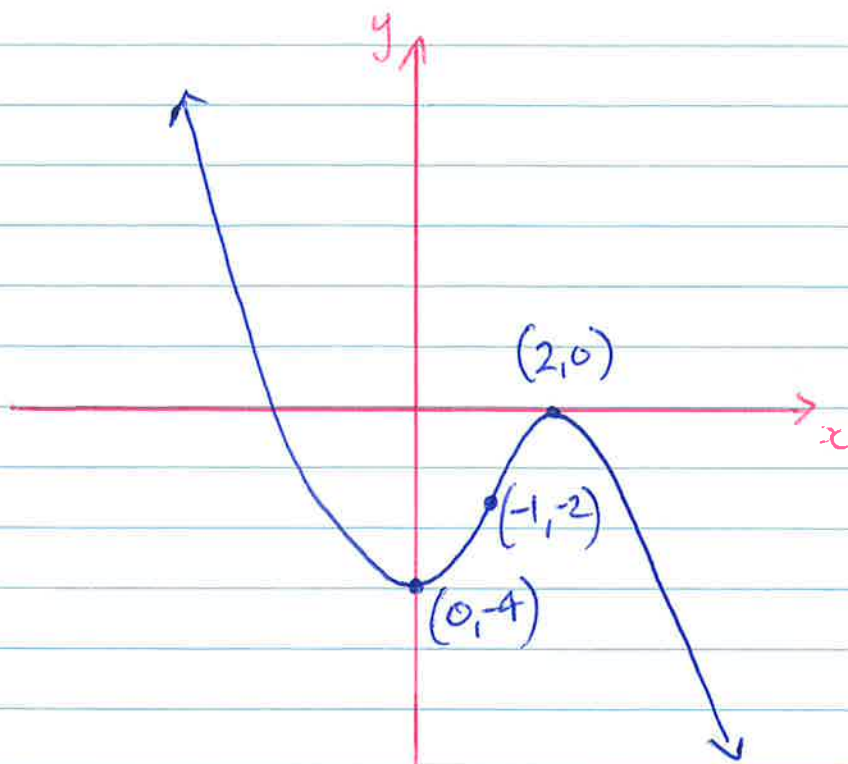
$$\text{at } x = 1 \quad y = -1^3 + 3(1)^2 - 4 = -2$$

x	0	1	2
y''	6	0	-6

$\therefore$ change in concavity

$\therefore (1, -2)$ is point of inflexion

iii)



b) i) $\frac{DE}{CD} = \frac{AB}{BC}$ (ratio of intercepts)

$$\frac{DE}{CD} = \frac{6}{4}$$

$$\frac{DE}{CD} = \frac{3}{2}$$

$$DE = \frac{3CD}{2}$$

ii) if $CD = 2$ then $DE = \frac{3(2)}{2} = 3\text{cm}$

In $\triangle BCD$ & $\triangle FED$

$$\frac{DE}{CD} = \frac{3}{2}$$

$\angle CDB = \angle EDF$ (vertically opposite angles)

$$\frac{DF}{DB} = \frac{7.5}{5} = \frac{3}{2}$$

$\therefore \triangle BCD \parallel \triangle FED$ (two matching sides in same ratio & included angle equal)

$$\begin{aligned}
 c) i) P_1 &= 140000 & 2018 \\
 P_2 &= 140000 \times 1.05 & 2019 \\
 P_3 &= P_2 \times 1.05 & 2020 \\
 &= 140000 \times 1.05 \times 1.05 \\
 &= 140000 \times 1.05^2
 \end{aligned}$$

ii) Note 2030 is P_{13}

$$\text{Total profit} = P_1 + P_2 + P_3 + \dots + P_{13}$$

$$= 140000 + 140000 \times 1.05 + 140000 \times 1.05^2 + \dots + 140000 \times 1.05^{12}$$

$$= 140000 (1 + 1.05 + 1.05^2 + \dots + 1.05^{12})$$

$$a = 1, r = 1.05, n = 13$$

$$S_{13} = \frac{1(1 - 1.05^{13})}{1 - 1.05}$$

$$= 17.7129 \dots$$

$$\therefore \text{Total Profit} = 140000 \times 17.7129 \dots$$

$$= 2479817.59 \dots$$

$$= \$2479818$$

Question 15

$$\begin{aligned}
 (15) \text{ a) } \int_1^e \frac{x^2+1}{x} dx &= \int_1^e \frac{x^2}{x} + \frac{1}{x} dx \\
 &= \int_1^e x + \frac{1}{x} dx \\
 &= \left[\frac{x^2}{2} + \ln x \right]_1^e \\
 &= \left(\frac{e^2}{2} + \ln e \right) - \left(\frac{1^2}{2} + \ln 1 \right) \\
 &= \frac{e^2}{2} + 1 - \frac{1}{2} \\
 &= \frac{e^2 + 2 - 1}{2} \\
 &= \frac{e^2 + 1}{2}
 \end{aligned}$$

$$\text{b) } 2\cos^2 x - 7\sin x + 2 = 0$$

$$\cos^2 x = 1 - \sin^2 x$$

$$\therefore 2(1 - \sin^2 x) - 7\sin x + 2 = 0$$

$$2 - 2\sin^2 x - 7\sin x + 2 = 0$$

$$-2\sin^2 x - 7\sin x + 4 = 0$$

$$2\sin^2 x + 7\sin x - 4 = 0$$

$$\text{let } u = \sin x$$

$$2u^2 + 7u - 4 = 0$$

$$2u^2 + 8u - u - 4 = 0$$

$$2u(u + 4) - 1(u + 4) = 0$$

$$(u + 4)(2u - 1) = 0$$

$$\begin{array}{l}
 P - 8 \\
 S \quad 7 \\
 F \quad 8, -1
 \end{array}$$

$$\therefore u + 4 = 0$$

$$u = -4$$

$$\sin x = -4$$

no solutions

$$2u - 1 = 0$$

$$u = \frac{1}{2}$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$c) y = (1 + x^2)^3$$

$$i) \text{ let } x = 0$$

$$y = (1 + 0)^3$$

$$= 1$$

$$\therefore A(0, 1)$$

$$ii) V = \pi \times \int_a^b [f(y)]^2 dy$$

$$y = (1 + x^2)^3$$

$$\sqrt[3]{y} = 1 + x^2$$

$$x^2 = y^{\frac{1}{3}} - 1$$

$$\therefore V = \pi \times \int_1^8 y^{\frac{1}{3}} - 1 dy$$

$$= \pi \times \left[\frac{y^{\frac{4}{3}}}{\frac{4}{3}} - y \right]_1^8$$

$$= \pi \times \left[\frac{3y^{\frac{4}{3}}}{4} - y \right]_1^8$$

$$= \pi \times \left[\left(\frac{3(8)^{\frac{4}{3}}}{4} - 8 \right) - \left(\frac{3(1)^{\frac{4}{3}}}{4} - 1 \right) \right]$$

$$= \pi \times \left(4 - \left(-\frac{1}{4} \right) \right)$$

$$= \frac{17\pi}{4} u^3$$

$$d) 2 \log_2(x+15) - \log_2 x = 6$$

$$i) 2 \log_2(x+15) - \log_2 x = 6$$

$$\log_2(x+15)^2 - \log_2 x = 6$$

$$\log_2 \frac{(x+15)^2}{x} = 6$$

$$\log_B A = P$$

$$\therefore 2^6 = \frac{(x+15)^2}{x}$$

$$64x = (x+15)^2$$

$$64x = x^2 + 30x + 225$$

$$0 = x^2 - 34x + 225$$

$$ii) x^2 - 34x + 225 = 0$$

$$(x-9)(x-25) = 0$$

$$P \ 225$$

$$S \ -34$$

$$F \ -9, -25$$

$$\therefore x = 9 \quad \text{or} \quad x = 25$$

Question 16

(16) a) i) $v = 3t^2 - 12t + 14$
 $v = 3(0)^2 - 12(0) + 14$
 $= 14 \text{ m/s}$

ii) $\Delta = b^2 - 4ac$
 $= (-12)^2 - 4 \times 3 \times 14$
 $= 144 - 168$
 $= -24$

for $v = 3t^2 - 12t + 14$

$a > 0$ & $\Delta < 0$

$\therefore$ positive definite

iii) $x = \int v$
 $= \int 3t^2 - 12t + 14 \, dt$
 $= \frac{3t^3}{3} - \frac{12t^2}{2} + 14t +$
 $= t^3 - 6t^2 + 14t + c$

at $t=1$ $x = 1^3 - 6(1)^2 + 14(1) + c$
 $= 9 + c$

at $t=3$ $x = 3^3 - 6(3)^2 + 14(3) + c$
 $= 15 + c$

$\therefore$ distance $= (15 + c) - (9 + c)$
 $= 6 \text{ m}$

$$b) x = 15 - 12e^{-\frac{t}{14}}$$

$$i) \text{ at } t=0 \quad x = 15 - 12e^0 \\ = 15 - 12 \\ = 3 \text{ cm}$$

$$ii) 10 = 15 - 12e^{-\frac{t}{14}} \\ -5 = -12e^{-\frac{t}{14}}$$

$$\frac{5}{12} = e^{-\frac{t}{14}}$$

$$\log_B A = P$$

$$\log_e\left(\frac{5}{12}\right) = -\frac{t}{14}$$

$$t = -14 \log_e\left(\frac{5}{12}\right)$$

$$t = 12.256..$$

$$= 12.3 \text{ days}$$

$$iii) x = 15 - 12e^{-\frac{t}{14}}$$

$$\frac{dx}{dt} = -12e^{-\frac{t}{14}} \times -\frac{1}{14}$$

$$= \frac{6e^{-\frac{t}{14}}}{7}$$

$$\text{at } t=8 \quad \frac{dx}{dt} = \frac{6e^{-\frac{8}{14}}}{7}$$

$$= 0.484..$$

$$= 0.5 \text{ cm/day}$$

$$\begin{aligned} \text{c) Area} &= 2xy + \pi x^2 + \frac{1}{2} \\ &= 2xy + \frac{\pi x^2}{2} \end{aligned}$$

$$\text{i) Perimeter} = \frac{1}{2} \times \pi \times 2x + 2x + 2y$$

$$80 = \pi x + 2x + 2y$$

$$2y = 80 - \pi x - 2x$$

$$y = 40 - \frac{\pi x}{2} - x$$

$$\therefore \text{Area} = 2x \left(40 - \frac{\pi x}{2} - x \right) + \frac{\pi x^2}{2}$$

$$= 80x - \pi x^2 - 2x^2 + \frac{\pi x^2}{2}$$

$$= 80x - 2x^2 + \frac{\pi x^2}{2} - \frac{2\pi x^2}{2}$$

$$= 80x - 2x^2 - \frac{\pi x^2}{2}$$

$$= 80x - x^2 \left(2 + \frac{\pi}{2} \right)$$

$$= 80x - \left(2 + \frac{\pi}{2} \right) x^2$$

$$ii) A = 80x - \left(2 + \frac{\pi}{2}\right)x^2$$

$$\frac{dA}{dx} = 80 - 2\left(2 + \frac{\pi}{2}\right)x$$

Max/min when $\frac{dA}{dx} = 0$

$$0 = 80 - 2\left(2 + \frac{\pi}{2}\right)x$$

$$2\left(2 + \frac{\pi}{2}\right)x = 80$$

$$\left(2 + \frac{\pi}{2}\right)x = 40$$

$$x = \frac{40}{\left(2 + \frac{\pi}{2}\right)}$$

$$x = 11.2019...$$

x	11	11.2	11.5
$\frac{dA}{dx}$	1.44	0	-2.12

/ — \

$\therefore$ Max area when $x = 11.2 \text{ m}$